

ST10440289

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Assignment 1

A. Introduction

CampusSwap is a mobile application that will assist students to exchange things in a very simple and convenient method. The application will enable users to exchange or sell second hand text books, clothing such as sportswear and specific electronics like calculators or headphones. The aim of this application is to help students save their money and also offer a viable platform where students can get access to cheap things.

Usability can be described as the ease and effectiveness in which a system can be interacted with by users. In CampusSwap, the usability is provided with an easy and straightforward interface to enable users to navigate the items, post their own items and request swaps or set up purchases with relatively little effort. An example would be where a user(student) can easily list an item by choosing a category and filling in a brief form with the required information. (GeeksforGeeks, 2026)

User experience (UX) on the other hand is the concern with the general feeling of the user using the application and the user's satisfaction. CampusSwap can improve UX by making sure to design a clean layout and easy navigation and feedback on completed actions. An example would be when a user manages to upload an item or even sends a request, a confirmation message will appear which makes the process more satisfying. (IBM, 2026)

The difference between usability and user experience is that usability's main focus is on how easy it is to use where as UX focuses on the overall users experience and satisfaction. They are both necessary in making CampusSwap effective and engaging for students.

B. User Centered Design (UCD)

User Centered Design(UCD) is a design methodology that aims at reviewing the needs, preferences and behaviors of the users during the development process. For my mobile app idea on CampusSwap, the main users are students who are searching for affordable ways to obtain or exchange items. (GeeksforGeeks, 2026)

UCD is important because it makes sure that the application is made to address the actual issues that were encountered by the students, like high prices of textbooks and availability of cheap clothes and electronics. Taking the user's needs in mind, the

application offers flexibility, allowing both swapping and cash-based options which is why the users are flexible and free to choose based on their needs. (GeeksforGeeks, 2026)

My application CampSwap is designed according to the principles of simplicity, consistency and visibility. To make the application easier to use and manage, it will be restricted to three types, namely books, textbooks, clothing such as sportswear and a selected electronics only. This will aid users to find relevant items fast without being overwhelmed.

Navigation is created in a bottom menu format and includes such sections like Home, add item, Messages and Profile. This is in accordance with the set standards of mobile design and it is simple to navigate the features. It uses clear labels and icons to enhance usability and eliminate confusion.

Using the principles of UCD, CampusSwap will make the application user friendly effective and focused on the needs of students.

C. Data Gathering Plan

Ways that data can be gathered is through the use of questionnaires and interviews to collect the user requirements of CampusSwap.

Questionnaires will allow for the collection of a large amount of data comprising of the students in an efficient manner. Using this approach will help determine typical preferences for example whether students would like to swap products or use cash and what kind of products they would like the most(textbooks,cloths or electronics). The questionnaires offer structured and simple to analyses information. (TechTarget, 2024)

Interviews provide additional information about the users experience and challenges faced. The interviews will allow students to describe certain problems like the difficulty to afford textbooks or to find reliable sellers. This assists in revealing the underlying needs that might not be reflected in questionnaires. (TechTarget, 2024)

The combination of both approaches will guarantee the balanced idea of user requirements. Questionnaires will give general information whereas interviews will give more detailed information, which justifies the design of a user centered application.

D. Design Principles.

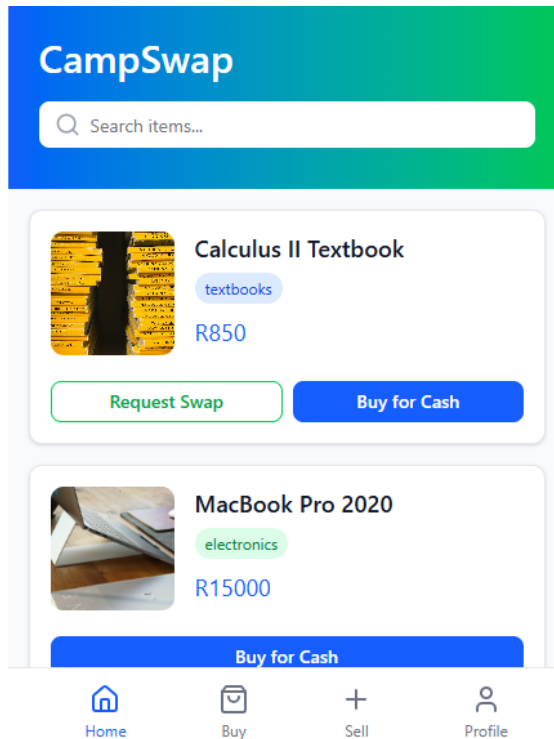


Figure 1 : Home Page (Figma, 2026)

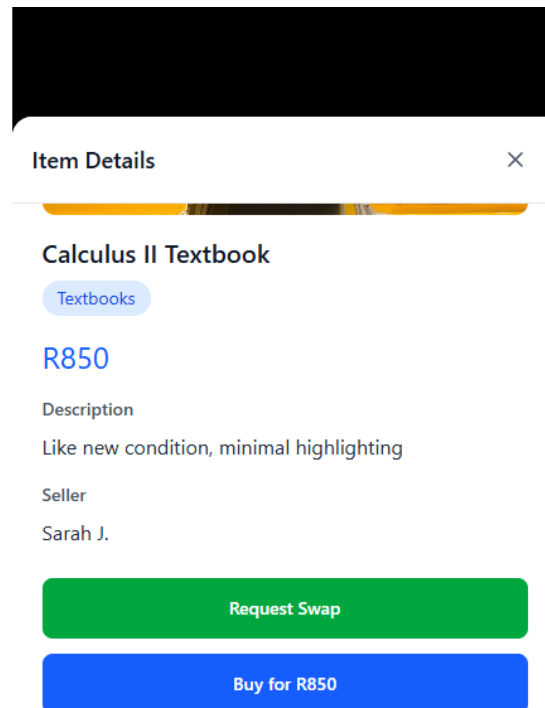


Figure 2: Item Details

1. Visibility

Visibility is the ability to show case relevant features and actions in a way that they can easily be noticed by the user to have an understanding of the available actions. (UX Design Institute, 2023)

Figure 1 shows the Home page of CampSwap, where items are listed and are properly shown with the main operations like request swap and buy for cash. Figure 2 shows page when you click on request swap or buy cash which allows the user to see more information about the item to be purchased. These buttons are highly prominently placed and can be recognized easily. Other than the category of item as like textbooks, clothing and electronics are well presented so that the user can rather easily understand the available options they have.

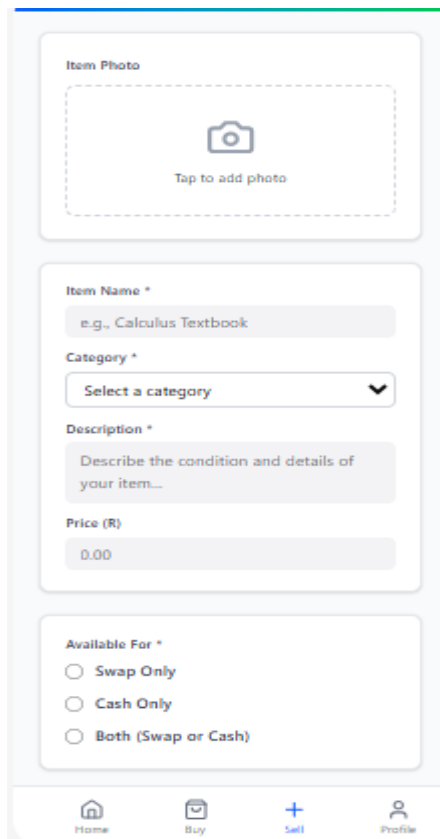


Figure 3: Sell page Interface (Figma, 2026)

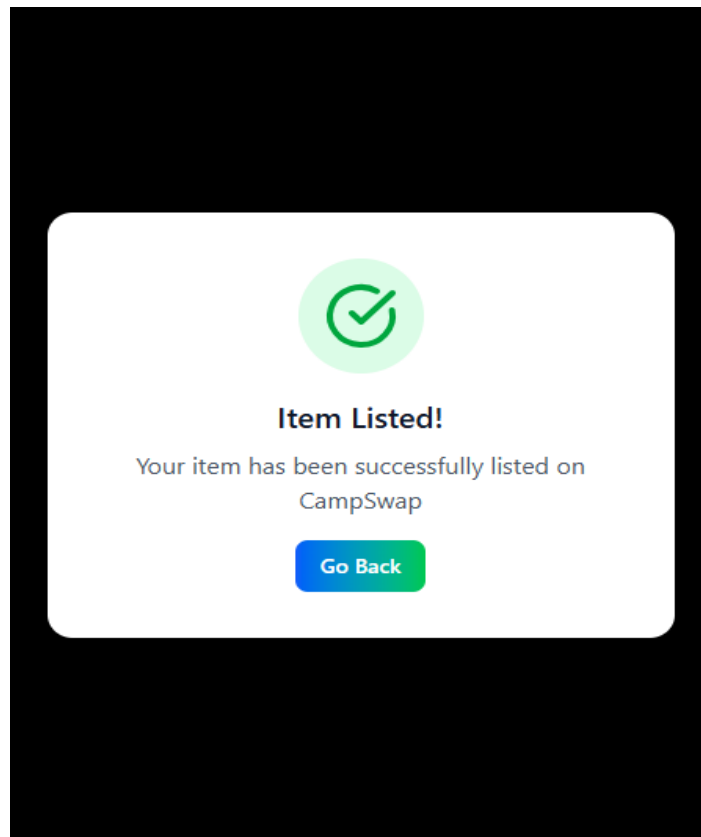


Figure 4: Confirmation Page

2. Feedback

Feedback can be described as a reply that the user receives after an action. (UX Design Institute, 2023)

The uploading of an item is also possible using the Sell Page as illustrated in figure 3 where one is required to fill in the appropriate details. In response to the submission of the item, the system issues feedback in a way that a confirmation message is given like item successfully listed as shown in figure 4. This will give the users confirmation that their action has been completed successfully and the general user experience is enhanced. (UX Design Institute, 2023)

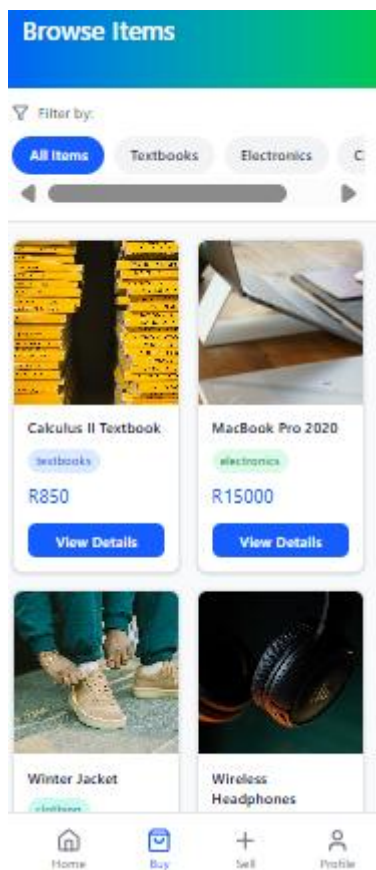


Figure 5 : Buy Page Interface
(Figma, 2026)

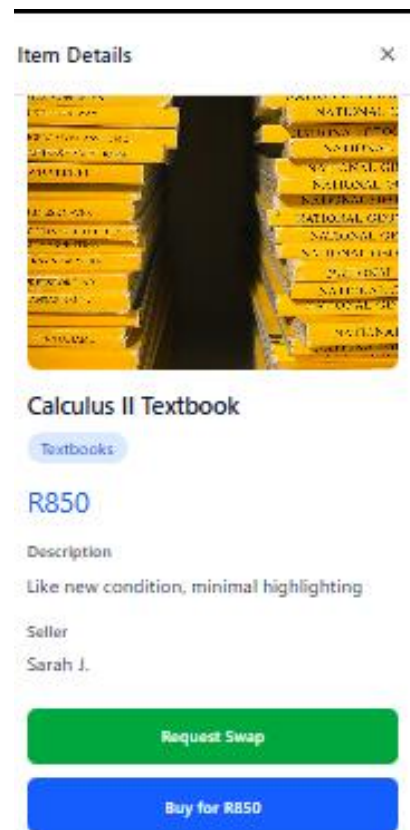


Figure 6: Buy Page view Details Interface

3. Consistency: Navigation across screens

Consistency is used to make sure that similar things are designed and act in the same manner throughout the application.

Referring to the buy tab the user is able to browse and view more information on certain item they are interested in thereafter once user views details their will be more information on the item with option to buy or swap.CampSwap maintains consistent layout, color scheme and structure of navigation with bottom navigation consisting of home, sell, buy and profile. The interface elements and buttons share the same design style and the application is easy to navigate. (UX Design Institute, 2023)

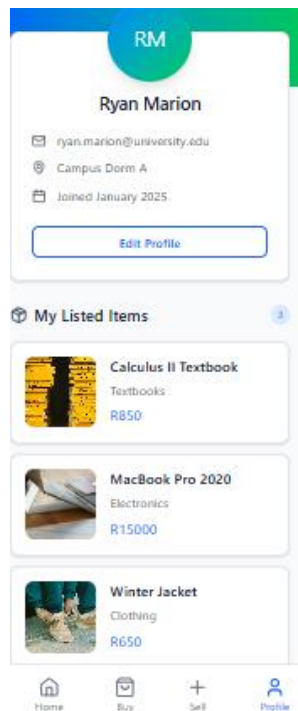


Figure 7: Profile Page Interface (Figma, 2026)

4. Affordance

The Concept of affordance is a design aspect where an object is crafted in such a manner that it is very clear to operate it. (Figma, 2026)

Figure 7 refers to profile tab where the user will be able to store their details and view list of items. Throughout the buy and sell tabs the buttons for example buy for cash are designed to be clickable making it clear for users to engage with them also input fields are used on the sell page figure 3 and 4, indicating where users can enter information. This helps the users understand the app making it less confusing. (Figma, 2026)

5. Error Prevention

Error prevention is aimed at reducing the risk of errors committed by users.

According to figure 3, the sell page require users to select a category of items, including textbooks, clothes or electronics before listing the item. This will then allow the user to make decision on whether an item is to be swapped or in cash or both. These are required fields that are used to discourage incomplete or incorrect submissions. (Figma, 2026)

E. Navigation and menus

CampusSwap navigation will be user friendly, easy to use and consistent everywhere. The bottom navigation bar helps users switch between the main parts of the application easily including Home, Buy, Sell and profile.

Its Home screen gives the user the ability to go through the available items by category and added items section gives the user the ability to upload items and it gives them option of whatever they can swap, purchase using cash or a combination of both. The messages section assists communication between the users to organize the exchanges, and the profile section stores users details and listing of the user. (Figma, 2026)

All menu entries are well marked and accompanied with icons to enhance the recognition. It uses the bottom navigation menu in keeping with the standard rules of mobile design and the menu is easy to use, and it makes cognitive load lighter. The regular positioning of the menu guarantees the ease of movements of the users in the application.

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